The Lyrical Time Capsule

Project Report

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Data Visualization for AI and Machine Learning

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Motivation

I chose this topic simply because I love music and was curious about how the mood and emotions in songs have changed over the last few decades. Many people argue that older music was happier or that modern songs have become more negative, and I wanted to see if the actual data supports this idea.

By analyzing song lyrics from the past 60+ years, my goal was to track shifts in overall sentiment, the main topics artists sing about, and how diverse the vocabulary has become over time. I also thought it would be interesting to compare the US, which heavily influences global pop culture, with Switzerland, to see if Swiss music trends followed the same path or showed a unique regional pattern. Ultimately, this project was a great way to combine my personal interest in music with the data visualization tools I learned during the semester.

Exploratory work

To conceptualize the analytical framework and the visual storytelling approach for this project, I drew heavy inspiration from the visual essay "Are Pop Lyrics Getting More Repetitive?" published by the data journalism site The Pudding[1].

The Pudding is a digital publication famous for its visual essays. In this specific project, data journalist Colin Morris analyzed the repetitiveness of Billboard Hot 100 songs over several decades. His methodology for measuring unique vocabulary directly inspired my approach to calculating and visualizing the "Lexical Diversity" of songs. Furthermore, the minimalist style of the essay served as the primary visual inspiration for my own minimalist, high-contrast layouts.

[1] https://pudding.cool/2017/05/song-repetition/

Data Acquisition and Methodology

United States Data Sourcing

For the US market, data collection relied on an existing Kaggle dataset titled "Top 100 Songs & Lyrics By Year 1959 - 2023 (USA)"[2]. While this provided a solid starting point, the raw data was not perfectly clean. It required manual preprocessing to fix formatting anomalies and correct corrupted text blocks before it could be safely used for analysis.

Swiss Data Sourcing

Because no pre-existing dataset was available for the Swiss music charts, the dataset had to be built from scratch.

First, I developed a Python script that scrapes historical chart data directly from hitparade.ch. Since the Swiss chart website only contains only the artist and title but no lyrics, another script was created to scrape the lyrics from Genius. Due to rate limits and the sheer volume of data, this API extraction process took several hours.

The Genius API introduced a high number of faulty entries. Consequently, a massive amount of manual data cleaning was required to verify the titles and ensure the Swiss dataset was accurate and comparable to the US dataset.

AI-Driven Text Analysis

Once both datasets were finalized, they were processed through an automated AI analysis script. Three specialized, pre-trained models from Hugging Face were implemented to extract deeper insights from the lyrics:

- **Sentiment Analysis:** cardiffnlp/twitter-roberta-base-sentiment-latest[3] was used to determine the sentiment of the lyrics.
- **Emotion Detection:** j-hartmann/emotion-english-distilroberta-base[4] was applied to identify specific emotional undertones within the text.
- **Topic Classification:** facebook/bart-large-mnli[5] was utilized for zero-shot classification, allowing the script to categorize songs into primary themes.

[2] https://www.kaggle.com/datasets/brianblakely/top-100-songs-and-lyrics-from-1959-to-2019
[3] https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest
[4] https://huggingface.co/j-hartmann/emotion-english-distilroberta-base
[5] https://huggingface.co/facebook/bart-large-mnli

Target audience and requirements

The primary target audience for my project are people with basic data-literacy but not necessarily music analytics expertise. The visualizations therefore prioritize clarity. Key requirements derived from this audience:

- **Low cognitive load:** single-message charts wherever possible, avoid encodings that require training to read.
- **Accessible color choices:** ensure palettes remain interpretable for common forms of color-vision deficiency.
- **Actionable annotations:** highlight specific songs or time periods that illustrate a trend, with short textual cues.

Design decisions

Overall Design Philosophy

To ensure the Data Story feels like a cohesive narrative rather than a disconnected collection of plots, a strict, unifying visual language was established and applied across all visualizations. The overarching design philosophy was guided by the reduction of cognitive load, consistent categorical encoding, and the maximization of the data-ink ratio.

Dark Theme

A custom dark theme was chosen as the foundation for the entire project, using a dark background paired with highly saturated, bright data colors. Because the background absorbs light, the brightly colored lines, bars, and heatmap cells visually pop and immediately draw the viewer's focus. This high-contrast approach looks modern and professional and significantly reduces eye strain when viewing digital reports.

Color Consistency

A fundamental rule of this Data Story was to never force the viewer to relearn the color legend. Throughout the charts, specific entities were permanently mapped to specific colors (e.g., the USA is consistently Blue, Switzerland is consistently Red). By maintaining this categorical color consistency, the viewer's brain builds a subconscious association after the very first chart.

Minimalist Design

To keep the focus entirely on the insights, a strict minimalist approach was applied across all charts, inspired by the principle of maximizing the data-ink ratio. All non-essential structural elements such as top and right borders, heavy gridlines, redundant axis labels, and boxed legends were deliberately omitted or faded into the background.

Chart 1: The Shifting Mood of Music

Choice of Chart Type

To visualize the development of sentiment over the years, a line chart was chosen, as it is the standard best practice for time series. Because sentiment scores can fluctuate heavily from year to year, I used a smoothed line (rolling average) as the main focus. The original yearly values remain visible in the background as slightly transparent points. This approach highlights the overarching trend without hiding the raw data.

Shaded Areas

Since the core message of this chart is the direct comparison between the two countries, the area between the two lines is shaded. The fill color changes dynamically depending on which country currently has a higher sentiment score. This immediately draws the reader's attention to the magnitude of the differences, making periods of convergence or divergence very easy to spot.

Contextual Annotations

Significant historical events were added as vertical markers. This helps the reader connect noticeable spikes or drops in the curves directly to real-world events. The text labels were placed alternatingly higher and lower so that they do not overlap and remain perfectly readable.

Chart 2: From Romance To More Dance

Choice of Chart Type

To visualize how the dominance of different song topics has shifted over time, a 100% stacked bar chart was chosen. This chart type is ideal for showing "part-to-whole" relationships across different time periods. The viewer can immediately compare the relative share of each topic per decade, making shifts highly visible.

Time Aggregation and Macro-Trends

Instead of plotting the data year by year, the timeline was aggregated into decades. Grouping the data into 10-year blocks smooths out short-term outliers and focuses the viewer's attention on long-term cultural shifts. This makes the overarching story much easier to grasp at first glance.

Reducing Cognitive Load

A crucial step in the data preparation was reducing the original eight sub-topics into three overarching categories. Human working memory struggles to track more than four to five categories (and colors) at once in a stacked chart. By consolidating similar themes (e.g., merging "Heartbreak" and "Romantic Love" into one category), the chart keeps the cognitive load low, allowing the core narrative to stand out.

Chart 3: Is Music Lossing It’s Vocabulary?

Visual Hierarchy and Layering

In this chart, a strong visual hierarchy was established. The individual songs are plotted as a background scatter plot with very low opacity. This pushes the raw data to the background. The 5-year smoothed median lines are plotted with a heavy line weight and solid colors, acting as the figure in the foreground. This allows the viewer to see the full variance and distribution of the data while immediately understanding the overarching trend.

Robust Aggregation

Instead of calculating the mathematical average for each year, the median was used. In music lyrics, extreme outliers are common. The median is a robust statistic that prevents these extreme outliers from artificially pulling the trendline up or down. Furthermore, applying a 5-year rolling window smooths out the remaining year-to-year noise, making the long-term gradual decline in lexical diversity much clearer.

Chart 4: Cultural Markers: What We Talk About

Choice of Chart Type

To visualize the frequency of different thematic categories across multiple decades, a heatmap was chosen. While a multi-line chart could technically display this data, lines often become tangled and difficult to read when dealing with several categories at once. A heatmap solves this by encoding values through color intensity, allowing the viewer to instantly scan the matrix for hotspots and overarching cultural shifts.

Data Normalization and Grouping

A critical design choice happened during the data preparation phase. Instead of tracking single, isolated words, related terms were aggregated into thematic buckets (e.g., "Phone" includes “call”, “mobile”, “text”, etc.). This prevents the data from becoming too sparse. Furthermore, the average mention per song were counted. This is important, because if we only plotted total word counts, decades with more songs in the dataset would unfairly dominate the chart.

Reducing Cognitive Load

While heatmaps are excellent for showing big-picture trends, it can be difficult for the human eye to accurately map a specific color back to an exact number on the legend. To solve this, direct labeling was applied, placing the exact average value inside each cell. This provides the best of both worlds: the colors show the macro-trend instantly, while the text provides micro-level precision without requiring the viewer to constantly look back and forth between the chart and the colorbar.

Work Table

Date	Hours	Task description
26.04	3.5	Exploratory work, initial US data sourcing, and setting up environment.
27.04	4.0	Writing web scraping scripts for Swiss charts, and Genius API integration for lyrics.
28.04	7.5	Extensive manual data cleaning, writing AI analysis scripts, and running Hugging Face Models.
29.04	7.0	Drafting visual concepts, setting up the configuration files (colors, fonts), and coding the initial Matplotlib/Seaborn charts.
05.05	8.0	Refining the visualizations: applying minimalist design, optimizing the data-ink ratio, and adding custom storytelling annotations.
06.05	6.0	Developing the interactive Data Story layout, integrating all charts, and finalizing the Streamlit application.
29.05	1.5	Fixing small bugs in the application and presenting the project to the class.
09.06	5.0	Structuring the academic report, writing the methodology section, and drafting the detailed design choice justifications.
10.06	3.5	Finalizing the report text, reviewing all visualization arguments against academic criteria, and formatting the final submission.
Total:	46.0	

Acknowledgment of Assistance

This report was written with the assistance of Gemini 3.1 Pro. All core ideas, insights, and design reflections are original to the author, Sven Betschart. The writing process involved dictating initial thoughts into Microsoft Word, using Gemini 3.1 Pro to refine and rephrase the draft, and conducting manual edits to ensure the final text accurately conveyed the intended meaning. The specific prompts utilized to develop the Data Story are documented within the main article, whereas the prompt used for this report is provided here:

"Correct any errors in this project report and rewrite it in a more academic way. The report shouldn't grow much bigger as it is. It should not exceed 11000 characters including spaces. Also, give me feedback and ideas on how I can improve the report."